

Week Summary

Introduction to HPC

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- Unit 1.1: What is HPC and what is it used for?
 - In HPC compute power is aggregated to achieve high performance
 - HPC is often used in physical sciences and engineering, and more recently life sciences
- Unit 1.2: The Top500 List
 - The Top500 list ranks supercomputers according to performance on the linpack benchmark
 - The list provides a way to track global trends in HPC since the mid 1990's
 - Fastest systems today achieve 10's-100's petaflops performance using over 1 million processor cores
- Unit 1.3: The need for parallelism
 - Individual processor cores are not getting significantly faster
 - We are still seeing an increasing number of cores per chip
 - Making use of parallelism and multiple processor cores is vital to achieve high performance

Week Summary

- Unit 1.4: The cluster architecture
 - Clusters are now the most popular HPC architecture
 - Clusters aggregate off-the-shelf resources to build cost-effective HPC systems
 - The main components are: compute nodes, interconnect, mass storage and login nodes
- Unit 1.5: HPC languages
 - HPC applications are usually written in a compiled language (C/C++ and/or Fortran) for speed
 - Parallelism usually needs to be added by the programmer
 - OpenMP and MPI are two common ways to used to parallelise programs written in C, C++ and Fortran

Remote log in

- The workshops are scheduled from week 1
- Access to the remote computer will be by SSH (secure shell)
- Instructions will be provided telling you how to log on from a local computer running Windows, MacOS or Linux